



Aerolytics — DaaS Intelligence Platform

Master Documentation & Shark Tank India Pitch Dossier | Version 1.3

Executive Summary: Aerolytics is an enterprise-grade Drone-as-a-Service (DaaS) AI platform combining sub-second YOLOv8 computer vision with interactive 3D WebGL Digital Twins across Energy, Agriculture, and Search & Rescue.

1. Executive Overview

Aerolytics converts raw aerial imagery into sub-second spatial intelligence across Energy (Solar PV), Agriculture (Crop Health), and Search & Rescue (Thermal SAR).

2. Problem & USPs

Problem: Traditional inspections cost ₹3,00,000+ per sweep, take days of manual walking, and expose workers to electrical & terrain hazards.

#	USP	Description	Impact
1	3-Pillar Domain AI	Purpose-trained YOLOv8 models for Energy, Agri, & Rescue	>91% mAP@50
2	Sub-Second Speed	GPU inference (~12-50ms) per image frame	100x Faster
3	3D Digital Twin	Spatial mapping pins 2D pixel defects into 3D WebGL point cloud	Exact Repair
4	Hardware Agnostic	Telemetry gateway parses feeds from DJI, Skydio, Autel, PX4	Zero Lock-In

3. System Features & Command Center

Status	Route	Capabilities
Active	/dashboard	KPI overview, severity charts, quick action tools
Active	/missions/new	Core Workflow: Upload imagery → Run AI inference
Active	/missions/[id]	Interactive 3D Three.js Digital Twin with spatial anomaly pins
Active	/fleet	Drone hangar: flight hours, maintenance log modal
Active	/mission-planner	Autonomous waypoint planner: speed, altitude export

4. Prerequisites & Local Execution Guide



Demo Login: `http://localhost:3000/login` | Email: `operator@aerolytics.io` | Password: `demo1234`

5. AI Model Benchmarks

Pillar	Weights	Classes	mAP@50	GPU Latency
Energy	solar_best.pt (6.2 MB)	cover, crack, dust, normal	93.4%	18 ms
Agriculture	agri_best.pt (6.3 MB)	Blast, brownspots, leaf blight, rust	91.8%	22 ms
Rescue	thermal_best.pt (6.2 MB)	person	94.7%	14 ms

6. Shark Tank India Investor Pitch & Q&A (INR ₹ Focus) IN

Business Model & Monetization

- **SaaS Subscriptions:** Starter ₹40,000/mo | Pro ₹1,200,000/mo | Enterprise ₹4,000,000/mo.
- **Usage Fees:** ₹1,200 per MW (Solar) | ₹600 per Hectare (Agri).
- **Unit Economics:** ACV of ₹20 Lakhs/yr, LTV of ₹80 Lakhs, CAC of ₹3.8 Lakhs (**5.2x LTV:CAC**).
- **Gross Margins:** 85% SaaS margin, 75% blended margin.

Market Size & Pitch Ask

- **SAM (India):** ₹3,400 Crores across 70+ GW Solar PV & Precision Agri.
- **Pitch Ask:** ₹1.5 Crores for 5% Equity at ₹30 Crores Valuation.
- **Allocation:** ₹67.5 Lakhs (45%) R&D/Edge AI, ₹52.5 Lakhs (35%) GTM Sales, ₹22.5 Lakhs (15%) ISO/Security.